

Project Specification and Baseline Experiment

Group 11

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Project Focus: Exploring Communication Tradeoffs

Primary focus dimension.

Our primary focus dimension is **reliability**. In this project, reliability refers to how consistently the receiver can decode the transmitted information correctly in a backscatter communication link. It can be evaluated using metrics such as bit error rate (BER), packet error rate (PER), and received signal strength indicator (RSSI).

Theory review.

Reliability in a backscatter communication system is strongly affected by the physical relationship among the carrier emitter, the tag, and the receiver. One important factor is the **position of the tag**. Since the backscattered signal experiences a cascaded propagation path, the total attenuation depends on both the carrier-to-tag distance and the tag-to-receiver distance. Therefore, moving the tag to different positions can significantly change the received signal strength and, as a result, the decoding performance.

Another important factor is whether the **carrier, tag, and receiver are located on the same straight line**. The geometric arrangement affects how efficiently the receiver can capture the reflected signal. In some positions, the backscattered signal may be stronger and easier to decode, while in others the signal may be weaker due to propagation loss or less favorable reflection conditions. This may lead to non-linear changes in communication performance even when the distances change only slightly.

The **receiver bandwidth, frequency deviation, and data rate** also influence reliability. A wider receiver bandwidth allows more signal components to pass through, but it also admits more noise, which can reduce decoding accuracy. Frequency deviation affects how distinguishable the modulated signal is at the receiver; if it is not well matched to the system conditions, symbol detection becomes less robust. The data rate also plays an important role: a higher data rate shortens the symbol duration, which generally makes the system more sensitive to noise and timing errors, while a lower data rate usually improves robustness at the cost of throughput.

In addition, **environmental noise and interference** can reduce signal quality. Background RF interference, multipath propagation, and other uncontrolled effects in the environment may distort the received waveform or lower the signal-to-noise ratio. As reliability decreases, this is typically reflected by a higher **BER** and **PER**. At the same time, **RSSI** provides an indication of received signal strength, which is often related to communication quality, although a stronger RSSI does not always guarantee a lower error rate if interference is significant.

Initial Proposal

1. Hypothesis / Research Question

Our main hypothesis is that **communication reliability can be improved by adjusting physical-layer parameters** in the backscatter system. In particular, we expect that changing the **2FSK frequency deviation Δf** at the tag, the **receiver bandwidth**, and the **frequency offset introduced by the tag** relative to the carrier can affect how robustly the receiver decodes the signal.

More specifically, we hypothesize that there exists a suitable combination of these parameters that leads to **lower BER and PER** and more stable reception under the same experimental conditions.

The goal is therefore to investigate how sensitive system reliability is to these PHY settings and whether reliability can be improved through parameter tuning.

2. Proposed Approach

Our proposed approach is to experimentally vary a small set of PHY parameters and evaluate their impact on reliability. The main parameters of interest are the **2FSK frequency deviation** Δf used by the tag, the **receiver bandwidth**, and the **backscatter frequency offset** relative to the carrier frequency.

The motivation is based on communication theory. First, the frequency deviation Δf determines how far apart the two 2FSK symbols are in frequency. If the separation is too small, the receiver may have difficulty distinguishing the symbols, especially in the presence of noise. If the separation is more appropriate, symbol detection may become more reliable. Second, the receiver bandwidth influences the tradeoff between capturing useful signal energy and admitting unwanted noise. A bandwidth that is too narrow may distort or attenuate part of the signal, while a bandwidth that is too wide may reduce robustness by allowing more noise and interference. Third, the frequency offset introduced by the tag affects where the backscattered signal appears in the spectrum. A more suitable offset may help the receiver separate the desired signal from interference or leakage from the carrier.

This approach fits well with the provided platform because the system allows us to configure these PHY-related parameters while keeping the hardware setup unchanged. By modifying one parameter at a time, or testing selected combinations of parameters, we can study how each setting affects BER, PER, and RSSI in a controlled way.

3. Experimental Plan

To evaluate our hypothesis, we will perform a controlled set of experiments in which we vary selected PHY parameters and observe their effect on communication reliability. The main **independent variables** are the **2FSK frequency deviation** Δf of the tag, the **receiver bandwidth**, and the **backscatter frequency offset** relative to the carrier. These parameters will be adjusted across a number of predefined values while the rest of the system configuration is kept unchanged.

The main **performance metrics** will be **bit error rate (BER)**, **packet error rate (PER)**, and **RSSI**. BER and PER directly reflect the reliability of the communication link, while RSSI provides supporting information about the received signal strength under different parameter settings. By comparing these metrics, we can study whether certain PHY configurations improve decoding performance.

To ensure a **fair comparison**, the experiments should be conducted under the same hardware setup and environmental conditions as much as possible. This means using the same carrier transmitter, tag, and receiver, keeping the same transmission power and packet structure, and performing the tests in the same physical environment. In addition, only one main parameter should be varied at a time during the initial evaluation, so that its effect can be observed more clearly. After that, selected combinations of parameters can be tested to investigate whether joint tuning leads to further improvements in reliability.

This experimental plan allows us to test our hypothesis in a systematic and reproducible way. If a certain configuration consistently results in lower BER and PER under the same conditions, it would support the idea that PHY parameter tuning can improve the reliability of the backscatter communication system.

4. Tradeoff Discussion

Although PHY parameter tuning may improve reliability, it also introduces tradeoffs. For example, increasing the **2FSK frequency deviation** Δf may make the two frequency symbols easier to distinguish at the receiver, which can improve detection performance and reduce BER or PER. However, a larger frequency deviation also means that the signal occupies a wider frequency span.

The **receiver bandwidth** also involves an important tradeoff. A wider bandwidth makes it easier for the receiver to capture the desired backscattered signal without cutting off useful spectral components. However, it also allows more background noise and interference into the receiver, which

may reduce reliability. In contrast, a narrower bandwidth can reject more noise, but it may also attenuate part of the desired signal if the bandwidth is too small.

The **backscatter frequency offset** relative to the carrier must also be chosen carefully. If the offset is too small, the backscattered signal may stay too close to the strong carrier component, which can lead to **self-interference** or carrier leakage at the receiver. Increasing the offset can help separate the desired signal from the carrier and make reception cleaner. However, the offset should still remain within a suitable range for the receiver configuration and the overall system design.

Overall, improving reliability is a parameter balancing problem rather than a single-direction optimization. The purpose of our experiments is to identify a combination of Δf , receiver bandwidth, and backscatter frequency offset that provides better reliability under the given hardware and environment.

Initial Experiment: Baseline Characterisation

1. Experimental Setup

Our baseline experiment follows the setup described in the project specification. The **carrier emitter** and the **receiver** were placed at a fixed distance of **3 m**. The **tag** was then positioned at five different locations along the straight line between the carrier and the receiver.

The five tested tag positions were chosen according to the distance from the carrier:

- 0.5 m
- 1.0 m
- 1.5 m
- 2.0 m
- 2.5 m

All other system parameters were kept unchanged in order to ensure a fair comparison between positions. For each tag position, the tag transmitted data and the receiver recorded the received packets in a log file. Based on the received packet data, we evaluated three reliability-related metrics: **packet error rate (PER)**, **bit error rate (BER)**, and **received signal strength indicator (RSSI)**.

Metric formulas

The following formulas were used to compute the metrics from the recorded `position*.txt` log files.

1. Packet Error Rate (PER) Let seq_i denote the sequence number of the i -th received packet in chronological order. The number of missing packets between two adjacent received packets is estimated as

$$m_i = (seq_i - seq_{i-1} - 1) \bmod 256$$

The total estimated number of missing packets is

$$M = \sum_{i=2}^N m_i$$

The estimated total number of transmitted packets inside the captured window is

$$T = N + M$$

Then the packet error rate is defined as

$$PER = \frac{M}{T}$$

2. Bit Error Rate (BER) The BER calculation compares the received payload against the expected payload generated from the pseudo sequence number.

For one packet,

$$BER = \frac{\text{number of wrong bits}}{\text{total number of received bits}}$$

For one log file,

$$BER_{\text{file}} = \frac{\sum \text{wrong bits across packets}}{\sum \text{total received bits across packets}}$$

3. RSSI RSSI is summarized as the arithmetic mean over all valid packets in one log file:

$$RSSI_{\text{avg}} = \frac{1}{N} \sum_{i=1}^N RSSI_i$$

Notes

- CRC is ignored in this analysis.
- The PER above is an estimate derived from sequence-number gaps in the received log.
- The figures use tag position on the X-axis and show PER, BER, and RSSI in three shared-X subplots.

2. Results

Figure 1 shows the measured **PER**, **BER**, and **RSSI** as functions of the tag position. The experiment was carried out on a table **outside the lab**, with the carrier and receiver separated by 3 m and the tag placed at five positions between them. The three subplots share the same X-axis, which represents the distance of the tag from the carrier.

From the figure, both **PER** and **BER** vary with tag position, but the trend is not fully monotonic. The worst reliability is observed around the 1.0 m and 2.0 m positions, where both PER and BER are relatively high. In contrast, the 0.5 m and 2.5 m positions show better performance, with lower PER and BER. The RSSI values also change with position, although the variation in RSSI does not perfectly match the error-rate curves.

Overall, the figure shows that tag position has a clear impact on communication performance. However, the measured results from this baseline experiment do not fully follow the simplest theoretical expectation that performance should become worst when the tag is located near the middle between the carrier and the receiver.

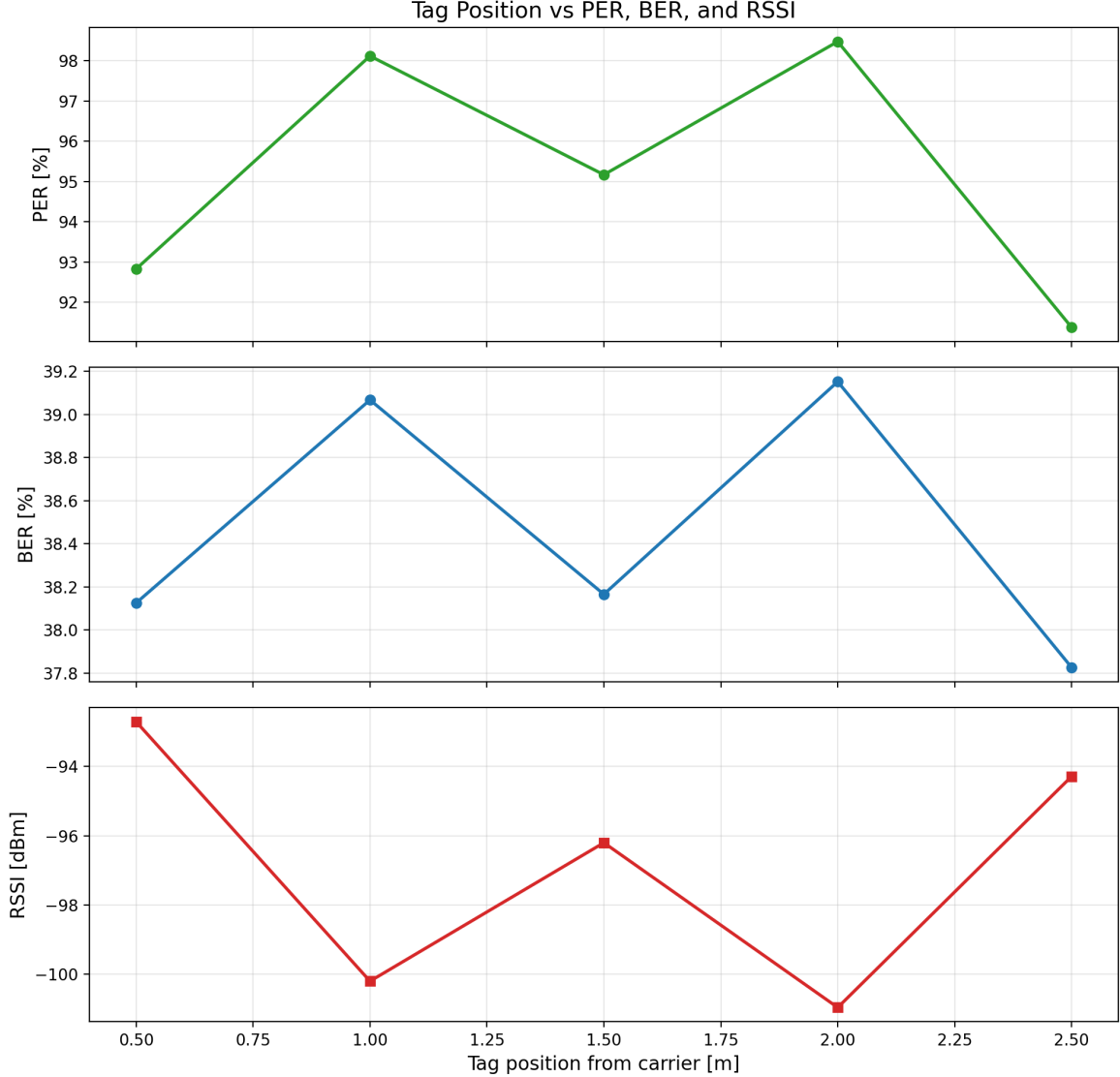


Figure. Three performance metrics versus tag position. PER is estimated from sequence-number gaps, BER is the bit-error rate, and RSSI is shown in dBm.

Figure 1: Measured PER, BER, and RSSI versus tag position in the baseline experiment. The carrier and receiver were placed 3m apart, and the tag was positioned at five locations between them. The experiment was performed on a table outside the lab, so the measured trends may have been influenced by external interference and environmental effects.

3. Discussion and Reflection

The results confirm that **tag position affects reliability**, but they also suggest that the real propagation conditions were more complicated than an ideal free-space model. In theory, when the tag is placed roughly in the middle between the carrier and the receiver, the cascaded backscatter path may lead to higher overall attenuation, and the communication performance could be expected to degrade. However, this pattern was not clearly observed in the outdoor-table experiment.

A likely explanation is that the experiment environment introduced additional uncontrolled effects. Since the measurements were performed **outside the lab**, the setup was more exposed to **external interference**, reflections from nearby objects, and multipath propagation. These factors can distort the simple distance-based trend and produce non-linear behavior. As a result, the position with the worst performance did not coincide exactly with the theoretically expected middle region.

Because of this inconsistency, we later repeated the experiment **inside the lab** under a more

controlled environment. In that follow-up experiment, we observed a trend that was more consistent with theoretical expectations. This suggests that environmental conditions played an important role in the baseline measurements and that the outdoor setup was not sufficiently controlled for drawing strong conclusions about the exact position-dependent behavior.

One limitation of this baseline experiment is therefore the **lack of environmental control**. In addition, only five tag positions were tested, which gives a useful first overview but may be too coarse to capture finer spatial effects. In a more controlled experiment, it would be better to reduce external interference, keep the surrounding geometry more stable, and possibly increase the number of tested positions or repeat each measurement multiple times to improve reliability and reproducibility.